

Author's response to reviewer's comments

Associate Editor and Reviewer comments:

AE, Dan Baldassarre: Your manuscript has now been reviewed by two experts, both of whom had many positive comments. However, both reviewers identified important issues that need to be addressed before your manuscript is suitable for publication in Ornithology. I won't rehash the reviewers' words too much, but to summarize what I see as important points:

- 1) Clearer discussion of the translocation and why it is important to your study design is needed.*
- 2) Many of your broader conclusions need to be tempered in light of small sample sizes, uncertainty about tracking spatial scale, and uncertainty about survival estimates.*
- 3) The relevance of your results on NOCA may not be broadly applicable to more human-sensitive species, so the big conservation/management implications should be tempered.*
- 4) Reviewer #2 mainly identified communication/clarity issues regarding the writing. Please take care to address those issues, which should also enable Reviewer 2 to offer more insight in a subsequent revision.*

Should you choose to submit a revision, you must address each reviewer point in a detailed response letter. Thank you for your submission!

Reviewer #1: Reviewer Report - ORNITH-26-00

General assessment

*This is a strong and timely manuscript addressing an important and under-studied topic: winter habitat selection and space use of the Northern Cardinal (*Cardinalis cardinalis*) in an urban context using telemetry. The field effort is substantial, the translocation design is creative and powerful, and the overall approach (tracking, home range estimation, RSFs across an urban gradient) is exactly the kind of work urban ecology needs more of. I also appreciate the transparency in the methods and the effort to connect results to applied questions.*

At the same time, several aspects of the manuscript require substantial revision. My concerns are not primarily about the fieldwork, but rather about clarity of study design, strength of inference, and degree of interpretation relative to what the data can support. In particular, the manuscript would benefit from:

- * Clearer communication of the translocation design (especially in the Introduction and Abstract),*
- * More cautious interpretation given small sample sizes and uncertainty in several analyses,*
- * Better acknowledgment of methodological limitations (e.g., spatial error vs. scale-of-effect claims, survival inference, correlated predictors), and*
- * Improved organization and clarity in several sections.*

I believe this manuscript has strong potential, and with careful revision it could make a valuable contribution.

We thank the reviewer for their insightful comments and critiques which have undoubtedly improved the manuscript greatly. We have provided further details below on how each of the major and specific comments below.

Major comments

1. Clarify the experimental (translocation) design in the Introduction and Abstract

A central strength of this study is that it uses a translocation design, not simply an observational comparison among sites. However, this is not clearly communicated in the Introduction or Abstract, which currently read as though birds were simply tracked across sites. This is a small change that would substantially improve clarity.

Because this affects how readers interpret inference, I strongly recommend:

** Explicitly describing this as a translocation experiment in the final paragraph of the Introduction.*

We thank the reviewer for this valuable insight. And have revised the last paragraph of the introduction to explicitly describe the nature of the translocation experiment to improve the clarity of this aspect of the study.

** Revising the Abstract to state clearly that birds were captured at a peri-urban site and experimentally released into suburban and urban sites. (see minor comments for wording suggestion)*

We have revised the abstract with the suggested wording the reviewer provided in their minor comment section below, with thanks!

This design is a major strength and should be foregrounded rather than obscured.

2. Strength of inference vs. sample size and uncertainty

Several key conclusions (e.g., larger home ranges in urban birds, mechanistic explanations for movement, survival differences) rely on small sample sizes for suburban ($n = 13$) and especially urban birds ($n = 9$), combined with large variance. While I appreciate the logistical difficulty of telemetry work and do not view this as a fatal flaw, I do think the manuscript currently overstates the strength of inference in places.

Across the Results and Discussion, I encourage the authors to:

** More clearly acknowledge uncertainty,*

** Avoid strong causal language where the data are correlative or underpowered, and*

** Frame mechanistic explanations more explicitly as hypotheses rather than conclusions.*

This applies especially to the home range comparisons, survival interpretation, feeder mechanisms, and scale-of-effect results. More information is including below in my minor comments.

We thank the reviewer for highlighting these shortcomings and have addressed all these issues throughout the paper. Specific changes are further detailed in the responses in the minor section comment below.

3. Scale-of-effect analyses and spatial precision

The manuscript places substantial interpretive weight on identifying 25 m vs. 50 m as "best" spatial scales. However:

** Predictors extracted from nested buffers (25, 50, 100, 200 m) are likely to be highly correlated.*

We thank the reviewer for this comment, and we agree that we should include correlation between the nested buffers. As we are not including variables from different scales within the same model, cross-scale correlations shouldn't inflate within-model multicollinearity. We highlight that the buffer comparison is more of a scale-selection exercise rather than a multi-

scale additive approach. This scale-of-effect approach (Martin and Fahrig 2012) is commonly used as a data driven approach to determine scale of landscape level effects in species-habitat models.

** Telemetry spatial error (~10-15 m) approaches the difference between focal scales (25 vs. 50 m). This makes fine-scale distinctions potentially unstable and overprecise.*

I strongly encourage:

** Reporting correlations among predictors across scales (or otherwise addressing cross-scale dependence),*

We thank the reviewer for this insight, we agree with the suggested addition and as a result, we have included a new paragraph to report the correlation between the buffer zones with more precision.

** Acknowledging explicitly that scale-of-effect results should be interpreted as "generally fine-scale" rather than precise distances, and*

We appreciate this comment and have, in accordance, added information for the reader to clarify that the results centered around scale-of-effect are not precise ecological thresholds.

** Tempering mechanistic or conservation claims that depend heavily on distinguishing 25 m vs. 50 m.*

We agree with the reviewer's comments and have, throughout the discussion, adjusted phrasing as to temper bold or over-confident claims.

4. Survival interpretation requires more caution

The apparent survival analysis is interesting, but the manuscript currently interprets differences as though they reflect true mortality patterns, when in fact most losses reflect missing individuals and the authors acknowledge possible emigration beyond monitoring areas.

These patterns are worth reporting, but should be framed more cautiously:

** Avoid implying strong causal conclusions (e.g., urban mortality drivers)*

** Present these mechanisms as plausible hypotheses*

** More clearly acknowledge uncertainty in apparent survival estimates*

We thank the reviewer for this critique and have sought to use language and include additional information to be more cautious in the framing of the survival interpretation throughout the paper. We provide more detailed information in the response to the minor comments below on this subject.

5. Broader generalization and conservation implications

Northern Cardinals are abundant, increasing, and highly synanthropic. They are a useful model for studying behavioral flexibility in urban environments, but they are not (likely) representative of conservation-sensitive urban species.

I encourage the authors to:

** More explicitly acknowledge this limitation, and*

** Avoid overgeneralizing management implications to urban biodiversity broadly.*

We thank the reviewer for this insightful comment. We have addressed this comment in the Implications section by tempering the statements made and rephrased sections about conservation-sensitive species to reflect the suggestions made.

Specific comments (by section)

Note to authors - this manuscript is built on excellent fieldwork and addresses a valuable question. My comments are aimed primarily at improving clarity, aligning interpretation with evidence, and strengthening the manuscript's conceptual rigor. With thoughtful revision, I believe this could become a strong contribution to the urban and avian ecology literature.

Abstract

** Line 11: "Habitat selection occurred at the smallest measurable scales (25-50 m)" is unclear biologically. A brief interpretive phrase would help clarify the ecological meaning (e.g., emphasis on microhabitat features such as vegetation structure or feeders).*

We thank the reviewer for pointing this out, we agree with this comment and have added a short section to clarify the ecological meaning.

** Line 19: The final sentence extends to broad conservation implications. Given that Northern Cardinals are abundant and highly synanthropic, consider tempering generality or framing implications as most relevant to urban-adapted species.*

We also agree with this comment. The last part of the highlighted sentence was revised to use a more cautious phrasing and focus the implications to urban-adapted species.

** Keywords: Include only terms not already in the title (e.g., "winter").*

We agree with this suggestion and have made changes accordingly.

** Abstract structure: The abstract undersells the experimental design. I suggest revising to clearly describe the translocation approach. For example (suggested wording, optional): "To better understand how birds use urban habitats during winter, we used VHF telemetry to quantify home range size, scale of habitat selection, and habitat associations in Northern Cardinals (*Cardinalis cardinalis*) experimentally translocated across an urban gradient in Montreal, Canada, over three winters."*

The authors thank the reviewer for this excellent suggestion and have made the suggested changes to the Abstract.

Lay summary

The lay summary closely mirrors technical language and may not quite be as accessible for lay people. I think it would benefit from more concrete phrasing (e.g., what birds actually did, where they moved, how they used feeders, why this matters). One way to approach this would be to write the lay summary as if explaining the study to a journalist preparing a short news piece: clear, simple, and focused on the main take-home messages.

The authors agree with this recommendation. We have worked on the Lay summary to apply the suggested changes in language to improve clarity and simplicity.

Introduction

** The Introduction is well cited but overly dense in places. Several paragraphs contain multiple distinct ideas without clear transitions.*

** For example, the opening paragraph could reasonably end at line 41, with the subsequent shift to habitat selection beginning a new paragraph with a clear topic sentence.*

** I encourage the authors to ensure each paragraph has a single clear idea and to use proper paragraph breaks for readability.*

We thank the reviewer for these three insightful comments on the structure of the Introduction. To resolve these comments, we added several paragraph breaks and made sure that each paragraph has an appropriate topic sentence. We reviewed the paragraphs to make sure they each contains a primary and well-developed idea with the goal of improving the reader's experience and the clarity of the work.

** Major structural issue: The final paragraph does not clearly convey the translocation design. This should be explicitly described here and in the Abstract.*

The authors agree with this comment from the reviewer that the translocation experiment should be described in greater detail in both the Abstract and Introduction. To resolve this comment, we added additional information at the end of the Introduction to make it clear that the study uses translocation as part of the study design, and have included the suggested wording by the reviewer in the minor comment above to the Abstract to improve clarity.

Methods

General: The overall telemetry, home range, and RSF framework is appropriate, but several clarifications are needed.

** Line 183: Tracking effort and spatial coverage differ substantially among sites (e.g., 5 km² vs 18 km²). Please clarify how effort was standardized or discuss this as a limitation.*

We thank the reviewer for pointing this out. We have now included in the manuscript and additional paragraph discussing how spatial coverage differences are mainly due to logistics and weather limitations, along with the time constraints of tracking birds (some individuals may only take 15 minutes to locate, while some may take up to 1h30 minutes). This makes standardizing the tracking efforts challenging, but as the overall goal was to cover as much area as possible to find an individual, standardization of effort was not central to field protocol.

** Line 185: "Cover as much ground as possible" is vague. Please provide more detail on effort (e.g., hours per visit, routes, number of observers).*

We agree with the reviewer's comment that clarity can be improved. In response, we have removed the vague statement and included more details on the process of tracking, including the hours spent tracking per session and the process of tracking newly relocated individuals, individuals with known locations and missing individuals. We hope that this improves clarity and precision to the manuscript while removing confusing information.

** Line 197: KDE estimates based on as few as five locations are unlikely to be reliable. Please justify this threshold, conduct sensitivity analysis, or consider excluding individuals with very low sample sizes.*

We employed the minimum threshold allowable by the analysis ($n = 5$) to make the most of a limited dataset. We agree that this may lead to oversmoothing and is less reliable than estimates made with more locations – but setting a more conservative threshold would have severely reduced our sample size. We endeavoured to use caution in the interpretations of our findings due to this logistical constraint in our analysis.

** Lines 198-208: The full mathematical formulation of KDE is unnecessary for this journal and could be replaced by a citation.*

Done - the mathematical formulation of KDE was removed and the citation from Worton, 1989 was added in its place

** Line 209: Clarify how low per-individual sample sizes may influence scale-of-effect analyses.*
We thank the reviewer for this comment. We have addressed this point in an earlier comment by adding a substantial section in the Scale-of-effect section to clarify the effect of low per individual sample sizes' effect on the analyses.

** Line 224: Some individuals contribute as few as five points to RSF analyses. Please justify this threshold or assess sensitivity.*

We are grateful for the reviewer's insight on this. We added a small section to highlight that even if an individual contributes to a small number of points, because we are comparing used/available at the population level, there are no minimum number of points required at the individual level.

** Line 232: Please justify the use of a 1:1 used-to-available ratio.*

We thank the reviewer for pointing this out. In response, we have added a clarification stating that the 1:1 ratio we used is based on the matched case-control design (e.g. Fortin et al. 2005), along with a short clarification sentence.

** Line 233: Please justify the 300 m buffer for available points.*

We agree with this comment and have included a section to add that the 300 m buffer comes from the median distance birds would travel/cover in a day based on tracking data.

** Line 235: Start a new paragraph when introducing models.*

The authors agree with this comment and have added a line return to divide the text.

** Models (general): The candidate model set appears hypothesis-driven, which is a strength. Consider explicitly framing these as a priori hypotheses.*

We agree that models were created for inference testing based on hypotheses and have further clarified this in the introduction and in the methods where the models are described.

** Line 250: Please provide actual VIF values or a correlation matrix among predictors. $VIF > 5$ suggests concerning collinearity, and this matters given the mechanistic interpretation offered.*
We thank the reviewer for this insight; we agree with the comment and have included the VIFs range for the predictor variables (between 1.05 and 3.31).

Results

** Lines 256-277 (Figures 2-3): These population trend and banding analyses feel tangential and interrupt the narrative. Consider condensing or moving to Supplement.*

The authors thank the reviewer for this comment. We have addressed this comment further in the document (3 pages below this present comment), under a large comment section regarding Figures 2, 3 and 6 which was initially included in the "Custom review questions". We have provided an elaborate response touching on all the concerns raised by the reviewer.

** Line 288-290: Delete "Table 1 details..." and instead reference Table 1 in the prior sentence.*
We thank the reviewer for this note; we have removed the unnecessary sentence and included a reference to the appropriate table in the previous sentence.

** Table 1: This table is difficult to interpret. The "fated survival" column (e.g., 90% for urban despite 17% apparent survival) requires clear explanation of assumptions.*
We thank the reviewer for highlighting this issue. As a result, we have decided to remove the confusing last column of the table as it isn't relevant for the rest of the manuscript and discussion. We hope that this improves clarity while making the table easier to read and understand.

** Table 2: Sample sizes (urban $n = 9$) and large SEs limit inference. Conclusions about urban effects on home range size should be tempered. Core area differences ($p = 0.077$) are better framed as a trend (done so on next line).*
We thank the reviewer for this insightful feedback. We have considerably tempered the conclusions made from the urban effects on home range size in the discussion section and generally throughout the paper. We agree with the reviewer that considering the low sample size and the significance only approaching significance, our conclusions must remain cautious.

** Line 324 / Figure 5: Scale-of-effect analyses rely on nested, likely highly correlated predictors. Please report correlations, clarify robustness, and temper interpretation. Consider moving Fig. 5 to Supplement.*
We agree with the reviewer's comment and have added correlations to increase the manuscript's transparency. As a result, we have also tempered the interpretations made in the discussion regarding scale of association to habitat. In addition, we have removed Figure 5 and have instead simply reported the Δ AICs that were used to determine the appropriate model. We hope that this solves this comment and improves readability of the paper along with conciseness.

** Line 352: Analytical procedure belongs in Methods, not Results.*
We have moved this part to the Methods section.

** Line 356 / Figure 6: Please more clearly articulate the ecological takeaway of this figure.*
We thank the reviewer for pointing out this lack of clarity. We have added additional information on page 19 to emphasize the ecological takeaways without including synthesis material that was reserved for the discussion section. We hope that this is a valuable addition for the reader improving the quality of this final section of the Results.

Discussion

** Line 379: The opening paragraph restates results rather than synthesizing contribution. Consider restructuring to goal → key findings → why they matter.*
We have re-written the opening paragraph of the Discussion accordingly and hope that the changes will reflect the suggested sequence the reviewer provided.

** Line 395: Survival interpretation should be more cautious; frame mechanisms as hypotheses.*

We thank the reviewer for this thoughtful comment; we agree and have adjusted several statements in the Lower apparent survival at urban site section to temper the causal language.

** Line 420: Feeder-home range mechanisms are plausible but speculative given sample size; temper causal language.*

We appreciate this comment by the reviewer. The highlighted section was significantly improved, particularly towards the end of the paragraph, to temper causal language and provide a more cautious view of the results and their implications. We also avoided to make claims regarding mechanisms as we recognize that this was not directly tested in our analyses.

** Line 439: Scale-of-effect interpretations are overconfident; acknowledge correlation and spatial error limitations.*

The authors thank the reviewer for the comment provided. We have worked on the scale of association part of the discussion to better acknowledge correlation and limitations, as well as tempered the causal language and the overconfidence that first transpired through the paragraph. We hope that these changes are adequate and present a more cautious interpretation of our results.

** Line 462: Several interpretations extend beyond data. Encourage clearer separation of results vs. speculation.*

We thank the reviewer for pointing this out as a key aspect to improve. We have worked on this section to add sentences that more clearly separate what comes out of our results vs. what is speculation regarding the potential explanations for those results. We have reframed multiple sections to be more cautious of our interpretations and provided a more realistic view of the results and their effects on cardinals' winter movements.

** Line 489: Cardinals are likely not representative of conservation-sensitive species; please acknowledge this limitation more explicitly.*

The authors agree with this comment from the reviewer. To resolve it, we have changed the Conclusion to better exhibit the limitation presented by the fact that cardinals are not conservation sensitive species.

Supplemental / Technical

** Line 726: Spatial error (10-15 m) approaches the difference between focal scales (25-50 m), which should be explicitly acknowledged in interpretation.*

We thank the reviewer for this comment. To resolve it, we have explicitly acknowledged the small gap between the spatial error and focal scales in the Discussion section in the Small scale of association to habitat regardless of urbanization level subsection.

** Graphical abstract: The current graphic implies hierarchical habitat selection not supported by the design. It should be revised to reflect associations tested rather than behavioral processes inferred. It also assumes migratory decision-making, as opposed to resident status, which is not demonstrated.*

We thank the reviewer for this comment. We have revised the graphical abstract to avoid conveying a sense of hierarchical habitat selection. We have also explicitly added the mention of "habitat association" to avoid confusion on what is being tested through the paper and clearly

define that this is not a behavioral process being tested. We have also changed the cardinal illustration to be a perched one to avoid the insinuation that migratory movement is at play here, in addition to the addition of a title highlighting the “winter” aspect of the research. We hope that these changes will improve the clarity of the graphical abstract.

****The following comments were pasted in by the AE and came from Reviewer #1 when answering the "Custom review questions". They may be somewhat redundant with their comments above, but I wanted to make sure the authors could see them. ****

Not all figures and tables are necessary in their current form. Figures 2 and 3 (regional population trends and banding-rate analyses) do not appear essential to addressing the core questions of the manuscript and interrupt the narrative focused on telemetry and habitat selection; these could be moved to the Supplement or substantially condensed. Figure 5 (scale-of-effect model ranking) appears largely procedural and offers limited standalone ecological insight; given concerns about cross-scale correlation among predictors, I suggest either moving this figure to the Supplement or summarizing the result more concisely in the text. Figure 6 (model prediction plots) would benefit from clearer articulation of the ecological takeaway; if the authors are unable to strengthen the interpretation, this figure could also be condensed or moved to the Supplement. The remaining figures and tables (e.g., study design, home range results, RSF coefficients) are generally appropriate and contribute meaningfully to the manuscript.

We thank the reviewer for this thoughtful suggestion. We agree that Figures 2 and 3 are not directly tied to the primary telemetry analyses, and we understand the concern that they may interrupt the narrative centered on habitat associations and space use. However, we feel that these figures provide valuable biological context that helps frame the study and interpret our results. Figure 2 illustrates the relatively recent northward expansion of the cardinals into southern Quebec, placing our study within the context of a population occupying the northern edge of the species' range. This context is relevant to understanding why winter ecology in this region is of particular interest. Figure 3 complements this by summarizing long-term banding data from our study site, highlighting seasonal patterns in local abundances and movement that provide background for interpreting the winter telemetry data. Together, these figures help situate our findings within both the historical expansion of the cardinals and the local ecological conditions at the study site. To address the reviewer's concern, we have substantially condensed both figures' accompanying text so that they provide context without significantly detracting from the main narrative. We believe this could be a good compromise that brings balance between maintaining the focus of the manuscript and providing readers with information that facilitates interpretation of the results we obtained. That said, we also recognize that opinions may reasonably differ regarding the appropriate level of contextual material in the main manuscript. If the editor or the reviewer feels that these figures are better suited to the Supplemental Material, we would of course be happy to move them there.

We have also added a few sentences to more clearly articulate the ecological takeaways of Figure 6 in the paragraph above the figure. We hope that this change in the manuscript will be enough to enhance the interpretations that can be extracted from the figure. If the editor and reviewer still feel that this figure should be reworked altogether, we will be happy to make more substantial modifications to the figure itself.

Several conclusions extend beyond what the data directly support and should be tempered. In particular, causal interpretations linking urban environments to reduced survival, larger home ranges, and specific mechanistic drivers (e.g., feeder density, vegetation structure, window collisions) are often presented more strongly than is justified given the small sample sizes (especially urban birds), high variance, and uncertainty around apparent survival (many individuals classified as missing rather than confirmed dead). These patterns are interesting and biologically plausible, but they should be framed more explicitly as hypotheses or tentative interpretations rather than firm conclusions. In addition, strong inferences about precise spatial scales of habitat selection (e.g., 25 m vs. 50 m) are not fully supported given likely cross-scale correlation among predictors and the positional error of telemetry locations. These results are better interpreted as evidence for generally fine-scale habitat association rather than support for a specific perceptual scale. Finally, some management and conservation implications are generalized too broadly. Because Northern Cardinals are abundant and highly urban-adapted, the findings are most applicable to synanthropic species and should not be extended to urban biodiversity more generally without qualification.

We thank the reviewer for this detailed feedback. All of the points mentioned in this paragraph were addressed through previous comments. Work was done on the language to bring it up to professional standard as well as to temper the conclusions and/or affirmations made with regards to the results we obtained. We have also paid special attention to the Implications paragraph to avoid over generalization and claims that extend beyond what the study covered. We hope that these modifications throughout the paper will resolve the vast majority of the points brought up in this paragraph.

Several aspects of the statistical approach raise concerns about robustness and interpretation. Home range estimates are based on very small numbers of fixes for some individuals (as few as five), which is unlikely to provide reliable KDE estimates and may bias comparisons among sites. Tracking effort and spatial coverage also differ among sites, yet it is unclear how effort was standardized or how these differences were accounted for analytically. The scale-of-effect analysis relies on predictors extracted at nested spatial buffers (25, 50, 100, 200 m), which are likely to be highly correlated. This makes model ranking across scales potentially unstable and increases the risk of spurious precision in identifying a “best” scale. Correlations among predictors across scales are not reported, and robustness of scale selection is not evaluated. In the RSF analyses, some individuals contribute very few locations, and the rationale for the 1:1 used-to-available ratio and the 300 m definition of availability is not clearly justified. These choices can meaningfully affect parameter estimates and inference. While the overall analytical framework is appropriate, these issues suggest that results should be interpreted more cautiously and that additional justification, sensitivity analyses, or transparency about uncertainty are needed.

The authors thank the reviewer for this relevant and clear comment. A section was added in the Non-breeding season tracking and home range estimations section to clarify and elaborate on the tracking efforts and procedure. We hope that this addition will resolve some uncertainty as to how tracking was conducted and the logistics issues making standardizing difficult (i.e. large discrepancies between tracking time of different individuals). We have also included the cross-scale correlations in the “Scale-of-effect” section to resolve the comment targeting the unreported correlations among predictors. We have also now included a section detailing the

rationale behind the use of the 1:1 ratio and the 300 m buffer zone. The 1:1 ratio is matched case-control (e.g., Fortin et al. (2005) while the 300 m was selected as this was representative of the median distance covered in a day by a cardinal based on tracking data collected. We hope that these modifications, along with the adjusted language to temper causal inferences will increase transparency of the manuscript.

Reviewer #2: *I greatly appreciated and enjoyed reading your manuscript and learning from it. The topic is important, and the field research comprehensively done and therefore may offer excellent quality information about northern cardinal ecology and avian urban ecology in general.*

For me, the issues that most stood out and are crucial to fix have to do with the writing quality, especially in the introduction and methods. The language vacillates between well-written and subpar, which I found disorienting. Similarly, some things are explained with great clarity, while others are cryptic and vague, especially in the methods. Make sure any reader with a higher scientific education can get a solid understanding of the logic and purpose of the technical and statistical methods you applied. Remember that some of these methods may be unfamiliar to most readers. The good news is that fixing these issues is very doable and will elevate the paper dramatically. I can't wait to read the improved version.

Here, I mostly avoided providing specific critical commentary on grammar or writing style, as suggested in the editor's letter. But, in certain cases where the text stood out to me as particularly problematic, I did provide comments - use these as examples of what do fix.

Lines 36 - 57: The focus on sparrows and especially ending the paragraph on the topic of sparrow, seems digressive. Also, there is some information from past studies about cardinal winter movement and range that would potentially be more relevant.

We thank the reviewer for this comment. We have gone back to this specific section and shifted the focus from sparrows to cardinals by including an older, but potentially more appropriate reference from Ritchison and Omer, 1990, detailing how Northern Cardinals form flocks in winter. We hope that this modification will enhance this paragraph of the Introduction section and resolve the comment in an efficient way.

Lines 48 - 55: The language does not meet professional standards.

We agree with this comment by the reviewer and have reviewed this section to include a language that meets professional standards.

Lines 58 - 78: The text about RSF's seems misplaced.

We thank the reviewer for this comment. In an attempt to resolve this, we have moved this section down in the text where we believe it eases the flow of the Introduction. We do think it's a valuable piece to keep in the Introduction to briefly discuss RSFs as they're a big part of the manuscript, but without going into too many details as this is done later on in the paper. We hope that this small change is enough to resolve the confusion around the RSF's paragraph placement.

Line 103 and/or 122-125: Weather conditions must be discussed accurately and with references.

** For example, "...region of extreme cold and loss of natural seeds during the cold winter months" : Referring to extreme cold must include quantitative data, for example a day/night*

averages, January temperature range, anything that can put a clear number on the adjective. Also consider a reference for the statement about seed loss.

We thank the reviewer for this insightful suggestion to improve the manuscript's precision. We have included additional clarifications on weather extremes in Montreal, including the January temperature range and a reference from the Government's Environnement Canada website regarding extreme cold weather conditions. We have also provided a new reference with regards to food availability; however, we want to note that the problematic sentence highlighted here stating "seed loss" was removed from the paper altogether. We hope that these changes are satisfactory to the comment made.

*After mentioning *C. cardinalis* for the first time in the text or each section, consider simply referring to the species as cardinal/s.*

We thank the reviewer for this comment. We agree with this and have made changes accordingly throughout the manuscript. We initially included *C. cardinalis* for the entire paper as it seemed to be a requirement from the journal, but if permitted we agree that simply using "cardinal" for the rest of the manuscript improves the readability of the paper.

Lines 114-116: Please rephrase this sentence because it is difficult to understand in its current form given that the subject in the sentence is the cardinals.

We thank the reviewer for pointing this out. We have rephrased the sentence as follow: "In addition, we predicted that locations with bird feeders, lower population density and a complex vertical vegetation structure would be associated with a higher probability of cardinals presence". This change shifts the subject of the sentence to the environmental variables instead of the cardinals, improving clarity.

Lines 190 - 191: ... "the probability density function of a random variable..." This phrasing is too cryptic. By 'random effect' did you mean the individual bird in this case? If so, phrasing it this way would make it clearer. Consider even simplifying the language further to truly give the reader a sense of what this calculation means. For example, to quote from Calenge 2023: "The home range deduced from the UD is the minimum area on which the probability to relocate the animal is equal to a specified value. For example, the 95% home range corresponds to the smallest area on which the probability to relocate the animal is equal to 0.95."

We thank the reviewer for this comment. We agree that this section could be clearer, as such we have rephrased the problematic section and have added additional sentences to improve clarity.

** For clarity and replicability, mention which function in the adehabitatHR R package you used, was it getverticeshr or kernelUD, something else?*

We thank the reviewer for pointing this out. We have added the function and package used to improve replicability of the analyses.

** Also, it seems that the Calenge CRAN document from April 2023 may also be informative as a reference here.*

We agree with this comment from the reviewer and have added the document as a reference.

** Lastly, the $K(u)$ formula you present here is similar but not the same as the one presented in Calenge 2023, which suggests to me that a theoretical reference, or a clarifying explanation about the formula, is needed here.*

We thank the reviewer for this comment. Previously, the other reviewer was suggesting removing it altogether from the manuscript as the presence of the formula does not bring anything to the understanding of the method or the analyses themselves. Instead of including the formula, we simply added a reference to it earlier in the text.

Lines 198 and 235: The former says the average was 17, the latter says 18.

We thank the reviewer for pointing out this conflicting information; 18 is the correct value and has been updated throughout

Line 199: Text needs to be rearranged, e.g. you'll need some text to say what the formula is about right before you show it, something like "We used the following formula to calculate....". This currently feels disjointed.

The authors are thankful for this comment by the reviewer. As mentioned earlier, we have removed the formula and as such resolved this issue.

Line 201: K is a probability, with all K 's adding up to 1. For clarity, it may be better to say 'probability' instead of 'weight' in ... "providing the weight assigned to a data point..." unless they are presented ranked. Also 'assigned' seems odd - aren't they simply calculated? Or is there some ranking system here?

We thank the reviewer, as above this has already been resolved with the removal of this section.

Figure 4 two boxplots side by side: The y-axes must align. Also, please use accurate language in the caption and throughout the results. For example, you need to state that these are home range log areas in the text, not simply "home ranges". Also, please explain in the caption what the points are. Mention that the plot shows boxplots as this carries a statistical meaning.

We thank the reviewer for the suggestions made to improve this figure. We have changed the figure so that the y-axes are now aligned and have improved the language in the figure caption to reflect the three main points mentioned by the reviewer regarding the log areas, the points description and the boxplot nature of the figure.